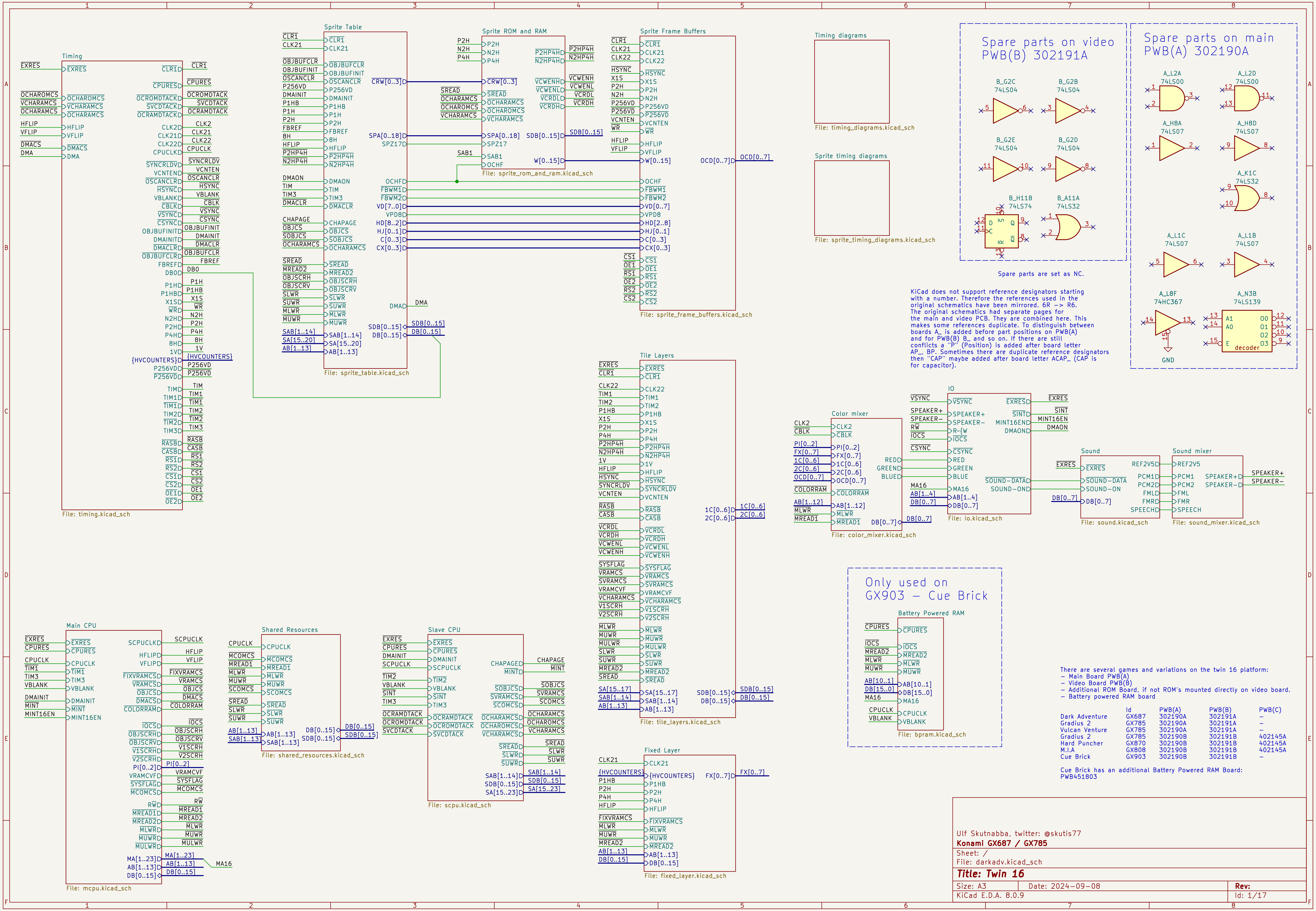


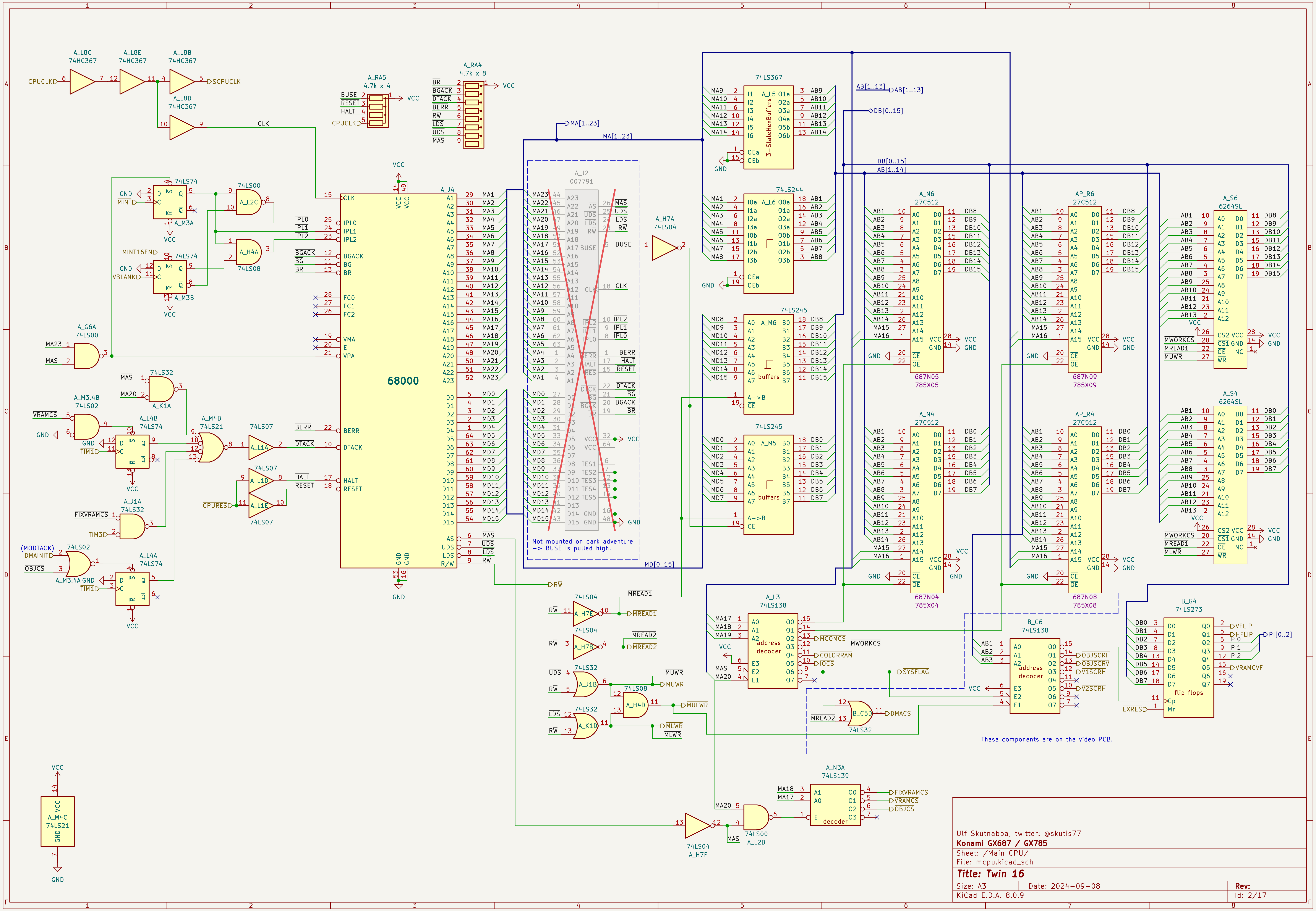
KiCad does not support reference designators starting with a number. Therefore the references used in the original schematics have been mirrored. 6R –> R6. The original schematics had separate pages for the main and video PCB. They are combined here. This makes some references duplicate. To distinguish between boards A_ is added before part positions on PWB(A) and for PWB(B) B_ and so on. If there are still conflicts a "P" (Position) is added after board letter AP_, BP_. Sometimes there are duplicate reference designators then "CAP" maybe added after board letter ACAP_ (CAP is for capacitor).

There are several games and variations on the twin 16 platform:
- Main Board PWB(A)
- Video Board PWB(B)
- Additional ROM Board, if not ROM's mounted directly on video board.
- Battery powered RAM board

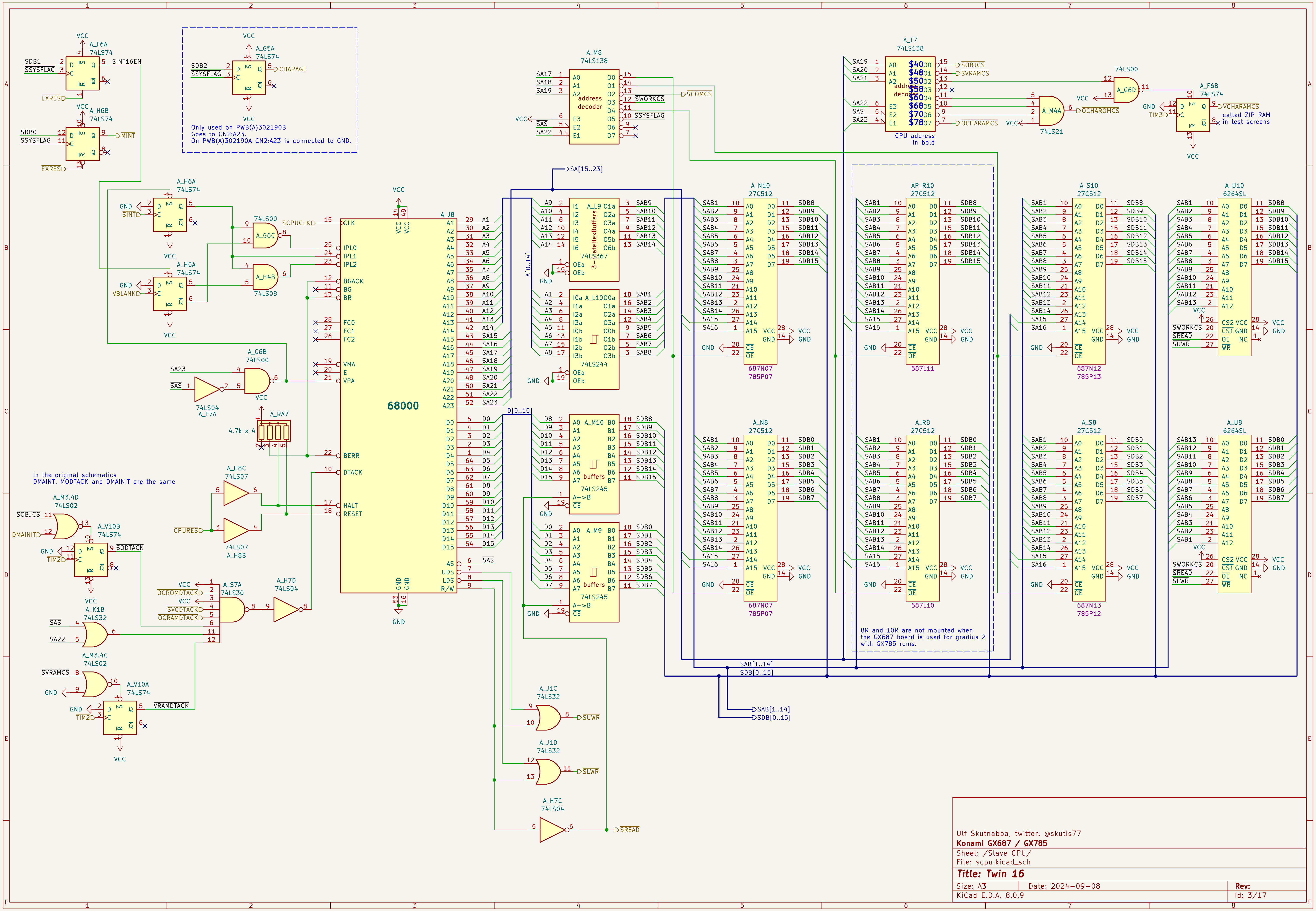
	Id	PWB(A)	PWB(B)	PWB(C)
Dark Adventure	GX687	302190A	302191A	–
Gradius 2	GX785	302190A	302191A	–
Vulcan Venture	GX785	302190A	302191A	–
Gradius 2	GX785	302190B	302191B	402145A
Hard Puncher	GX870	302190B	302191B	402145A
M.I.A	GX808	302190B	302191B	402145A
Cue Brick	GX903	302190B	302191B	–

Cue Brick has an additional Battery Powered RAM Board: PWB451803

Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /
File: darkadv.kicad_sch
Title: Twin 16
Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 1/17



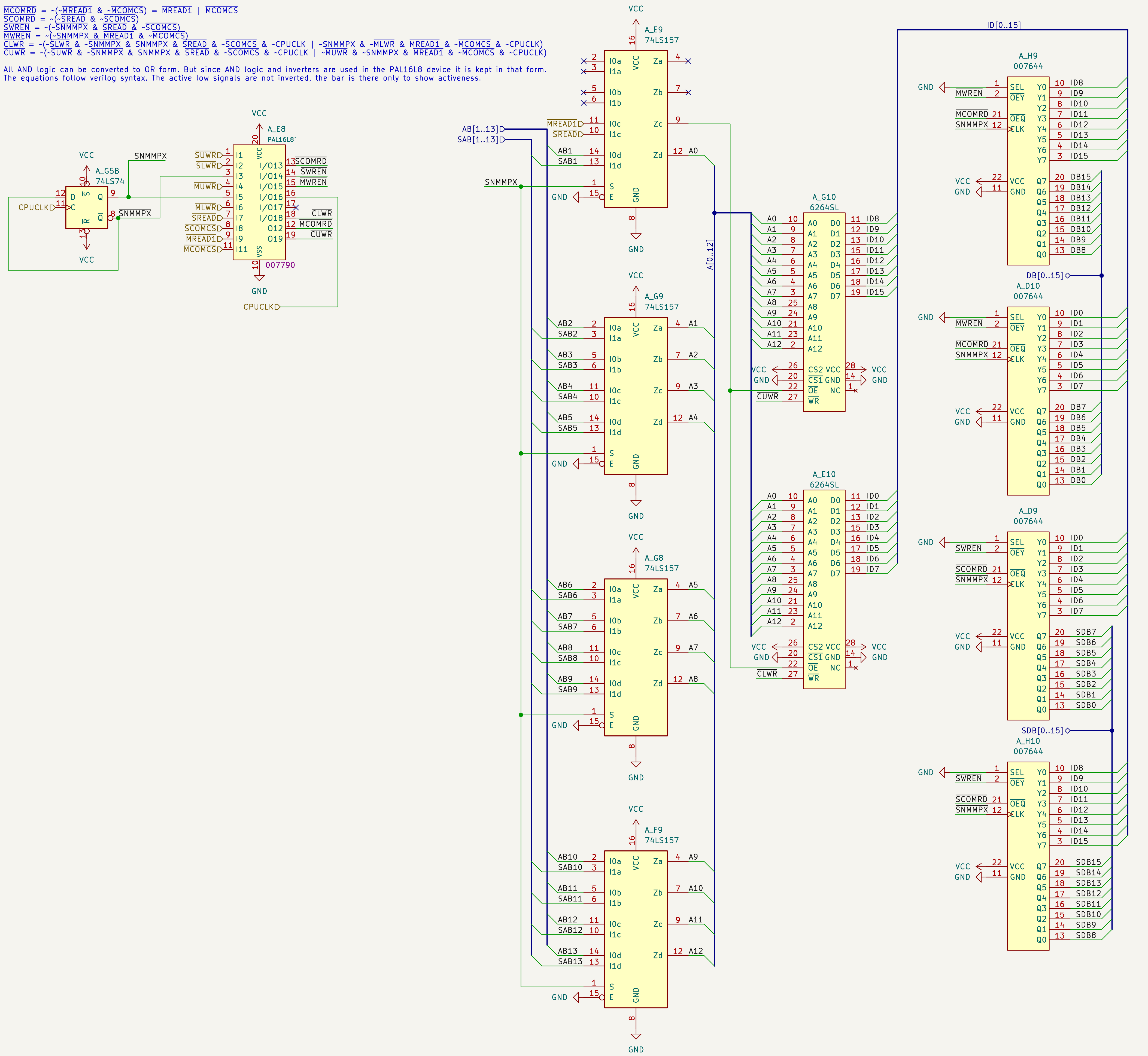
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Main CPU/
File: mcpu.kicad_sch
Title: Twin 16
Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 2/17


Only used on PWB(A)302190B
Goes to CN2:A23.
On PWB(A)302190A CN2:A23 is connected to GND.
In the original schematics
DMAINT, MODTACK and DMAINIT are the same
8R and 10R are not mounted when
the GX687 board is used for gradius 2
with GX785 roms.
CPU address
in bold
called ZIP RAM
in test screens
address
decoder
3-StateHexBuffers
buffers
68000
A_J8
A_F6A 74LS74
A_H6B 74LS74
A_G5A 74LS74
A_H6A 74LS74
A_H5A 74LS74
A_G6C 74LS00
A_H4B 74LS08
A_G6B 74LS00
A_F7A 74LS04
A_RA7
4.7k x 4
A_H8C 74LS07
A_H8B 74LS07
A_M3.4D 74LS02
A_V10B 74LS74
A_S7A 74LS30
A_H7D 74LS04
A_K1B 74LS32
A_M3.4C 74LS02
A_V10A 74LS74
A_M8 74LS138
A_L9 74LS367
A_L1000a 74LS244
A_M10 74LS245
A_M9 74LS245
A_J1C 74LS32
A_J1D 74LS32
A_H7C 74LS04
A_T7 74LS138
$40 $48 $50 $58 $60 $68 $70 $78
74LS00 A_G6D
A_M4A 74LS21
A_F6B 74LS74
A_N10 27C512 687N07 785P07
AP_R10 27C512 687L11
A_S10 27C512 687N12 785P13
A_U10 6264SL
A_N8 27C512 687N07 785P07
A_R8 27C512 687L10
A_S8 27C512 687N13 785P12
A_U8 6264SL
SA[15..23]
A[0..14]
D[0..15]
SAB[1..14]
SDB[0..15]
SINT16EN
MINT
CHAPAGE
SCOMCS
SWORKCS
SSYSFLAG
SOBJCS
SVRAMCS
OCHAROMCS
OCHARAMCS
VCHARAMCS
SODTACK
VRAMDTACK
SUWR
SLWR
SREAD
SCPUCLK
CPURES
EXRES
VBLANK
SINT
DMAINIT
TIM2
TIM3
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Slave CPU/
File: scpu.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.9
Id: 3/17

Konami 007790

```
MCOMRD = ~(~MREAD1 & ~MCOMCS) = MREAD1 | MCOMCS
SCOMRD = ~(~SREAD & ~SCOMCS)
SWREN = ~(~SNMMPX & SREAD & ~SCOMCS)
MWREN = ~(~SNMMPX & MREAD1 & ~MCOMCS)
CLWR = ~(~SLWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~SNMMPX & ~MLWR & MREAD1 & ~MCOMCS & ~CPUCLK)
CUWR = ~(~SUWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~MUWR & ~SNMMPX & MREAD1 & ~MCOMCS & ~CPUCLK)
```

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form.
The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.



This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs.
If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

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Konami GX687 / GX785

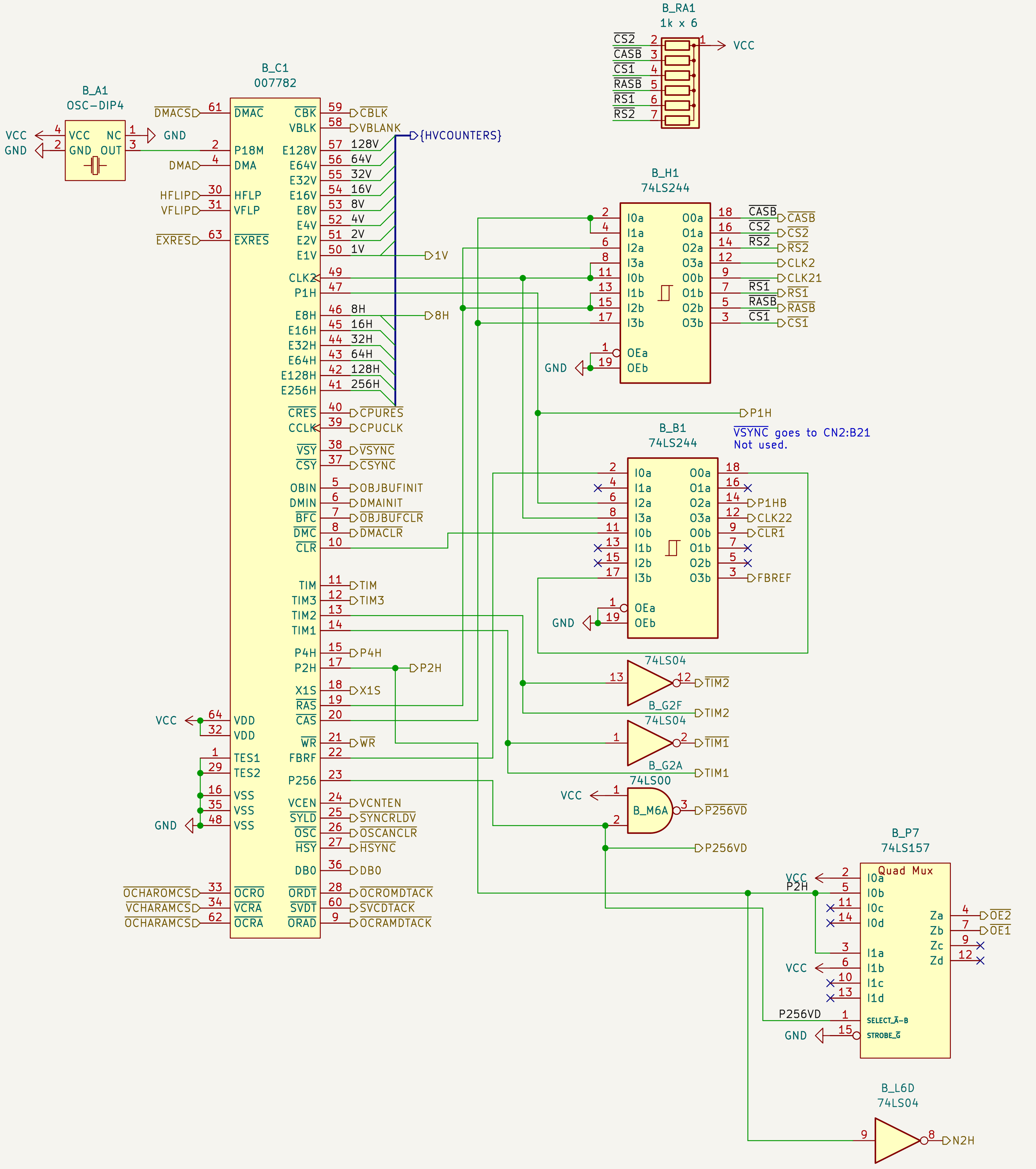
Sheet: /Shared Resources/

File: shared_resources.kicad_sch

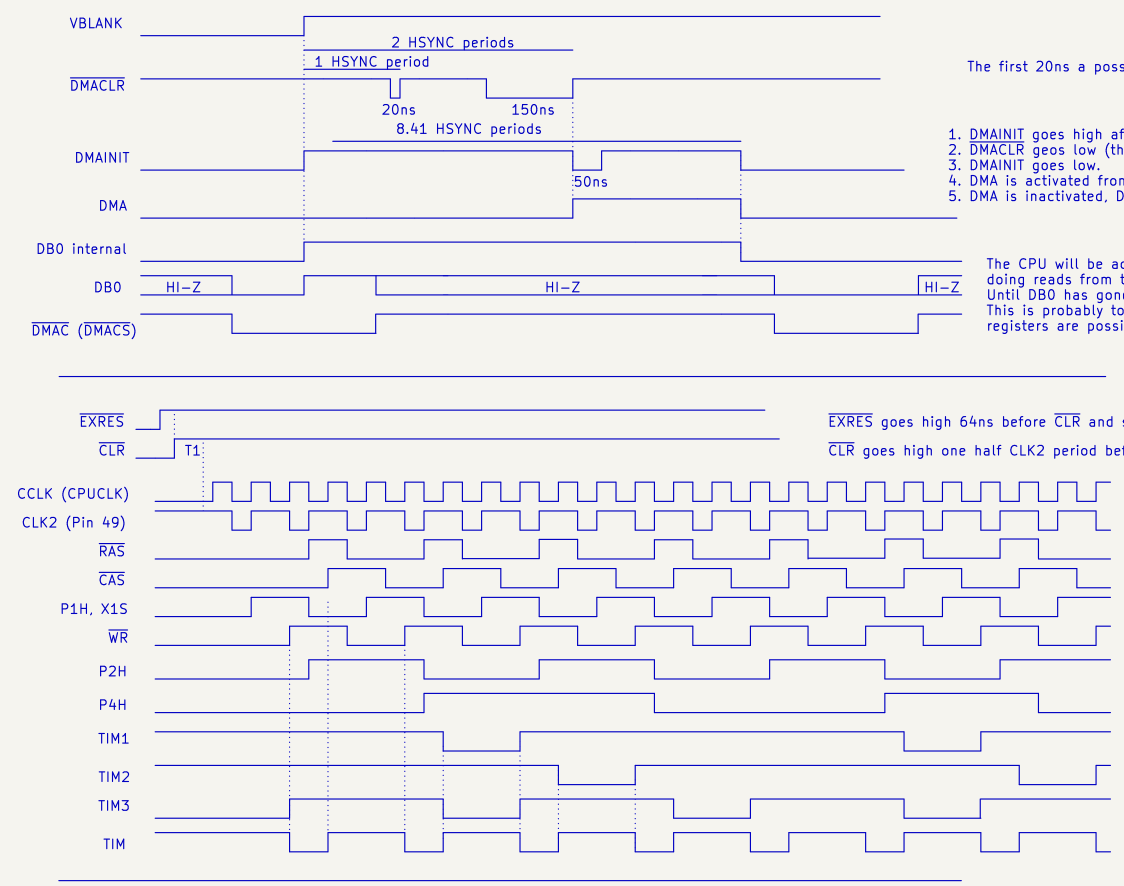
Title: Twin 16

Size: A3 | Date: 2024-09-08 | Rev:

KiCad E.D.A. 8.0.9 | Id: 4/17



DMA Synchronization



The first 20ns a possible glitch?

1. DMAINIT goes high after VBLANK goes active
2. DMACLR geos low (the second time) and high after 150ns.
3. DMAINIT goes low.
4. DMA is activated from the 00783(Sprite engine).
5. DMA is inactivated, DMAINIT, DB0 internal goes low.

The CPU will be activating regularly DMACS, doing reads from the system registers periodically. Until DB0 has gone from low to high and low again. This is probably to see if writes to the sprite controller registers are possible.

EXRES goes high 64ns before CLR and stays high.

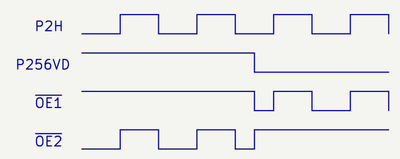
CLR goes high one half CLK2 period before the first falling edge of CLK2.

CCLK = 18.432MHz / 2 = 9.216MHz
CLK2 = 18.432MHz / 3 = 6.144MHz,
T = 1/f = 162.76ns, T = 2 x T1
f = CLK2 / 2 = 3.072MHz, T2 = 3 x T1

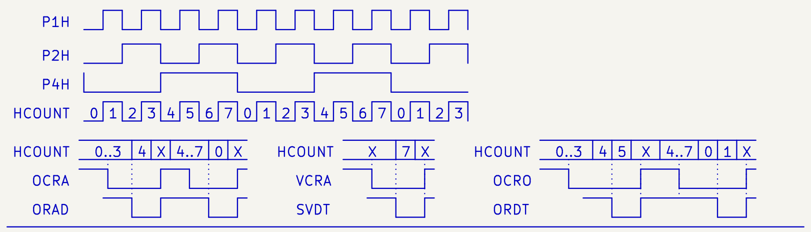
f = CLK2 / 2 = 3.072MHz
f = CLK2 / 2 = 3.072MHz
f = CLK2 / 4 = 1.536MHz
f = CLK2 / 8 = 768kHz
f = CLK2 / 8 = 768kHz
f = CLK2 / 8 = 768kHz
f = CLK2 / 4 = 1.536MHz
f = CLK2 / 2 = 3.072MHz

Timer	Usage
TIM1	Main CPU Tilemap RAM access
TIM2	Slave CPU RAM Tilemap RAM access
TIM	Sprite timing/address generator 007783
TIM3	Sprite timing/address generator 007783

Output enables active vs. part of frame buffer active



Chip select signals and Data Acknowledge signals



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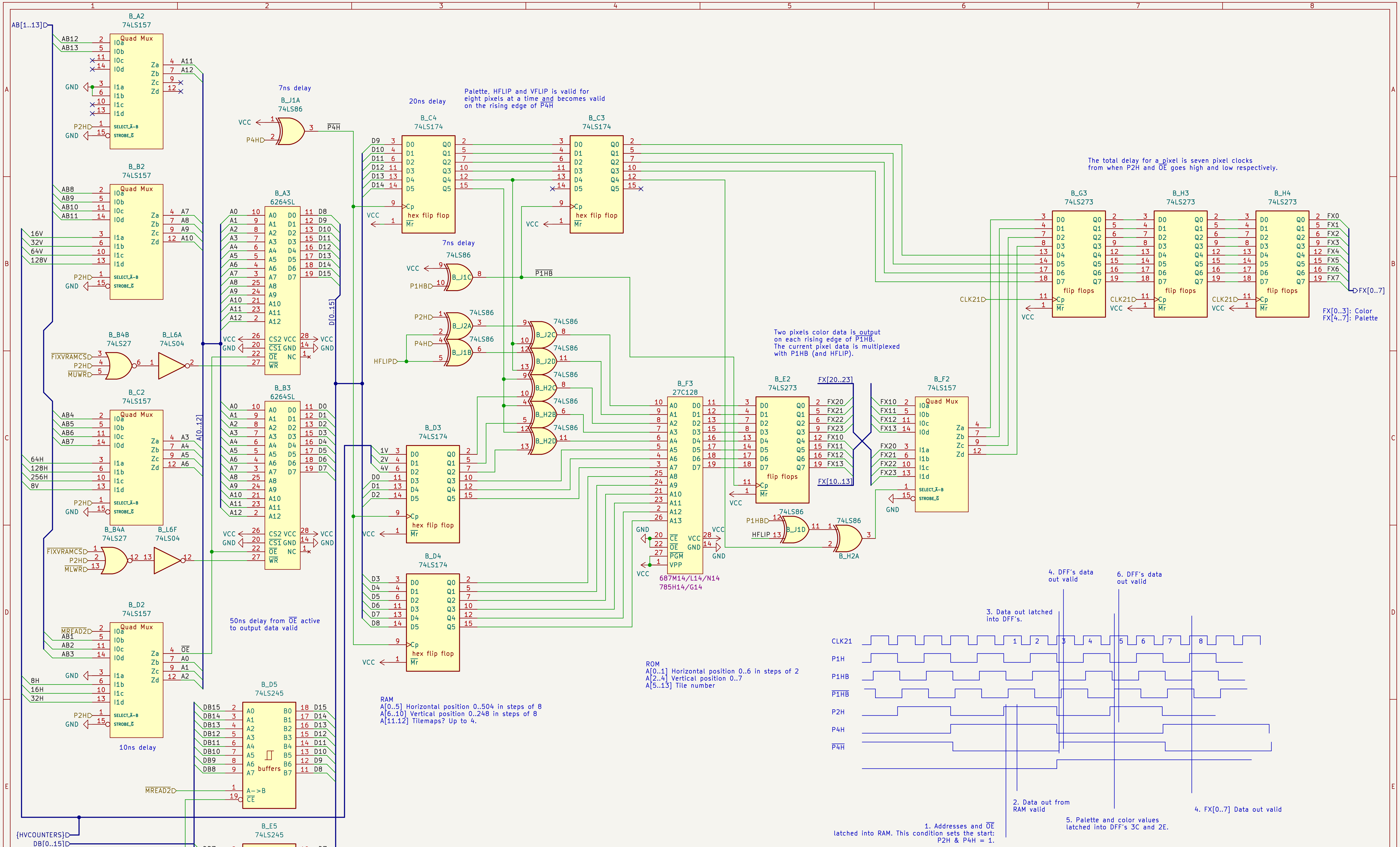
Konami GX687 / GX785

Sheet: /Timing/
File: timing.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 5/17



Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

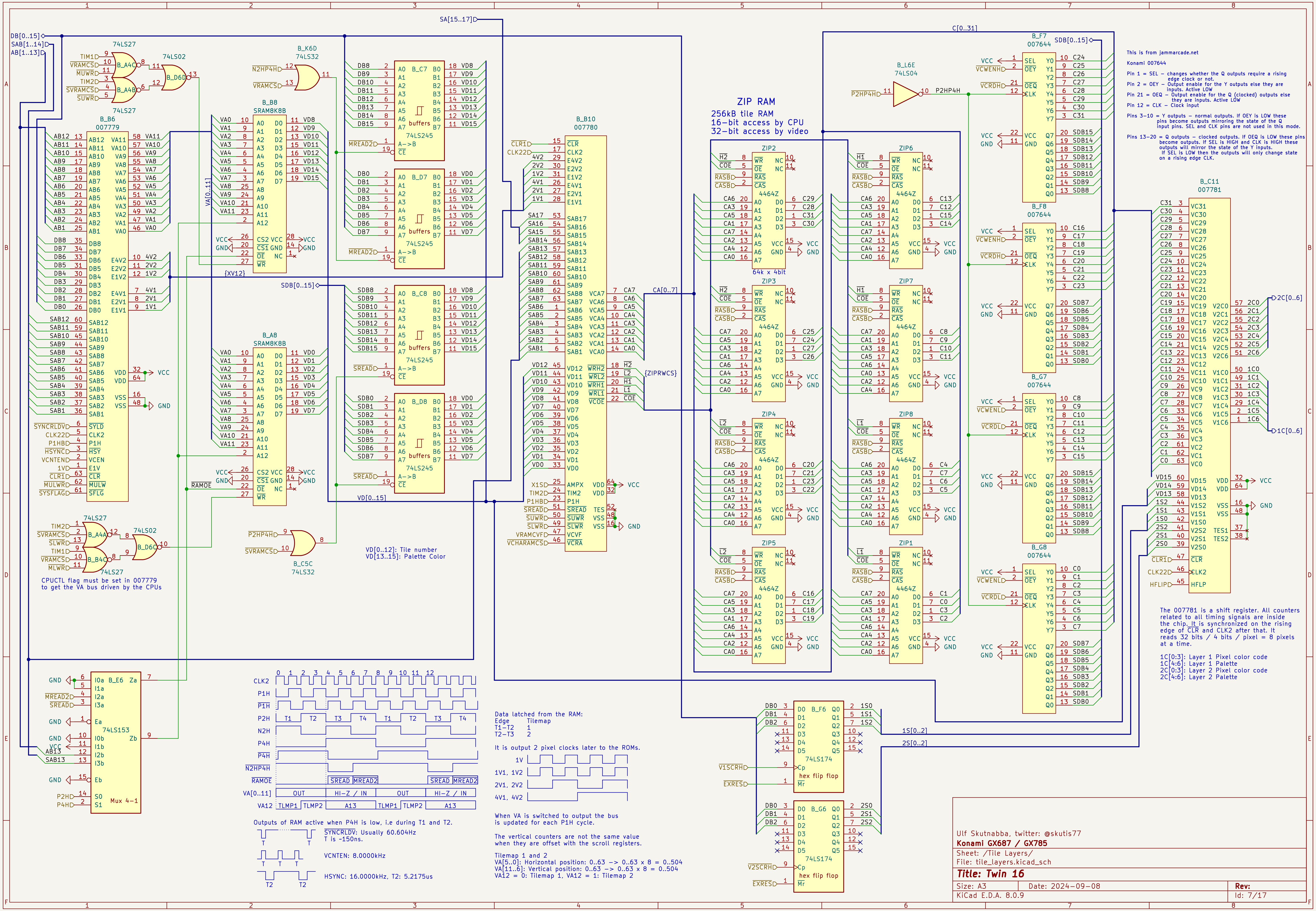
Sheet: /Fixed Layer/

File: fixed_layer.kicad_sch

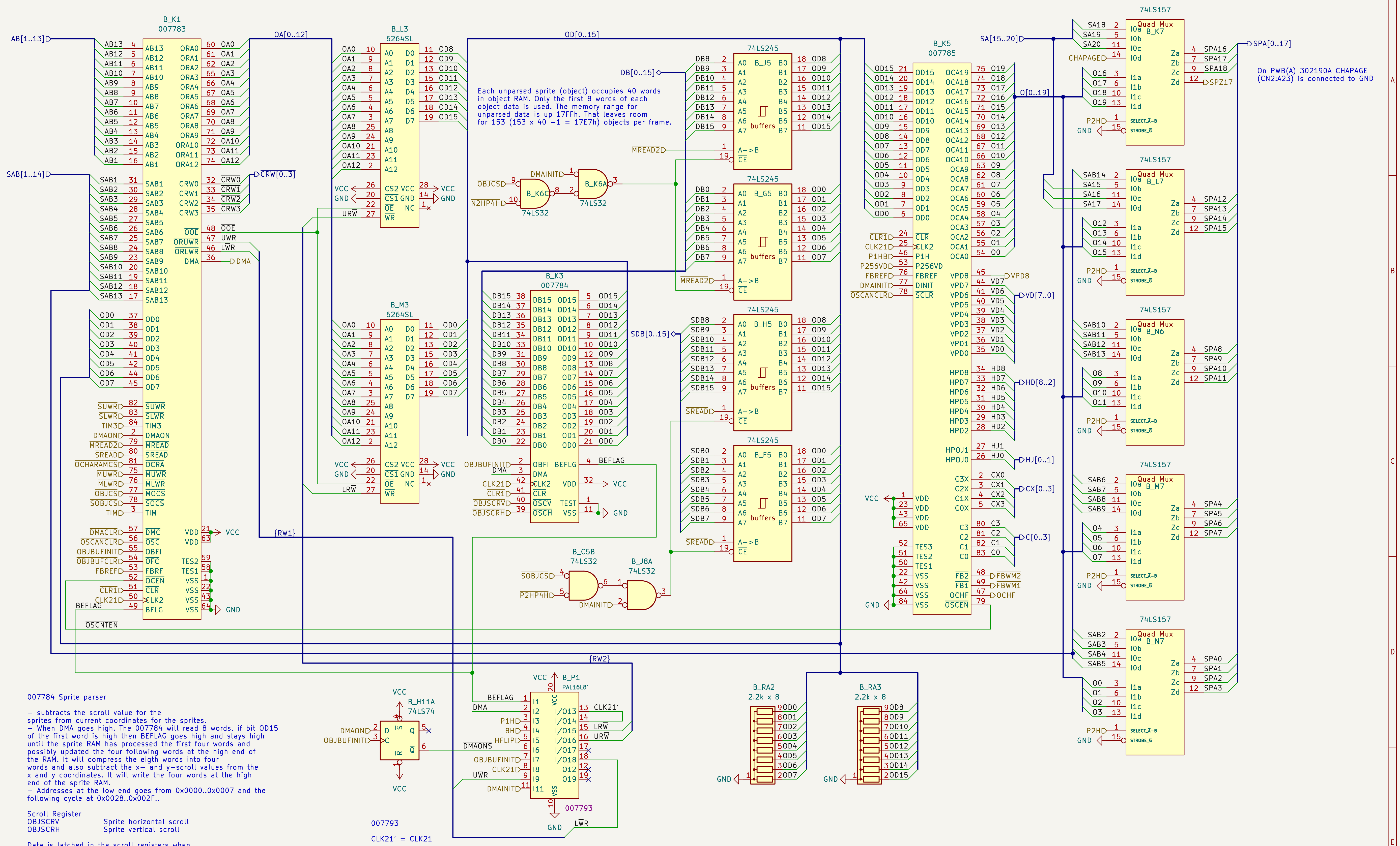
Title: Twin 16

Size: A3 | Date: 2024-09-08 | Rev:

KiCad E.D.A. 8.0.9 | Id: 6/17



ZIP RAM
256kB tile RAM
16-bit access by CPU
32-bit access by video
74LS27
74LS02
B_K6D
74LS32
B_B8
SRAM8K8B
B_C7
74LS245
buffers
B_D7
B_B6
007779
B_B10
007780
B_A8
SRAM8K8B
B_C8
B_D8
B_C5C
74LS32
B_E6
74LS153
Mux 4-1
ZIP2
ZIP3
ZIP4
ZIP5
ZIP6
ZIP7
ZIP8
ZIP1
4464Z
64k x 4bit
B_L6E
74LS04
B_F7
007644
B_F8
007644
B_G7
007644
B_G8
007644
B_C11
007781
B_F6
74LS174
hex flip flop
B_G6
74LS174
hex flip flop
CPUCTL flag must be set in 007779
to get the VA bus driven by the CPUs
VD[0..12]: Tile number
VD[13..15]: Palette Color
This is from jammarcade.net
Konami 007644
Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input
Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.
Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.
The 007781 is a shift register. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that. It reads 32 bits / 4 bits / pixel = 8 pixels at a time.
1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette
CLK2
P1H
P2H
N2H
P4H
N2HP4H
RAMOE
SREAD
MREAD2
VA[0..11]
OUT
HI–Z / IN
VA12
TLMP1
TLMP2
A13
Outputs of RAM active when P4H is low, i.e during T1 and T2.
SYNCRLDV: Usually 60.604Hz
T is ~150ns.
VCNTEN: 8.0000kHz
HSYNC: 16.0000kHz, T2: 5.2175us
Data latched from the RAM:
Edge Tilemap
T1–T2 1
T2–T3 2
It is output 2 pixel clocks later to the ROMs.
1V
1V1, 1V2
2V1, 2V2
4V1, 4V2
When VA is switched to output the bus
is updated for each P1H cycle.
The vertical counters are not the same value
when they are offset with the scroll registers.
Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 -> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 -> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Tile Layers/
File: tile_layers.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.9
Id: 7/17



Each unparsed sprite (object) occupies 40 words in object RAM. Only the first 8 words of each object data is used. The memory range for unparsed data is up 17FFh. That leaves room for 153 (153 x 40 −1 = 17E7h) objects per frame.

007784 Sprite parser

– subtracts the scroll value for the sprites from current coordinates for the sprites.
– When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eigth words into four words and also subtract the x– and y–scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
– Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV   Sprite horizontal scroll
OBJSCRH   Sprite vertical scroll

Data is latched in the scroll registers when the signals $\overline{\text{OBJSCRV}}$ and $\overline{\text{OBJSCRH}}$ go low.

007793

CLK21' = CLK21

$\overline{\text{LRW}}$ = ~($\overline{\text{DMAONS}}$ & OBJBUFINIT & CLK21' |
BEFLAG & DMA & P1H & $\overline{\text{DMAONS}}$ & CLK21' & (~8H & HFLIP | 8H & ~HFLIP)
~LWR & ~DMAINIT)

$\overline{\text{URW}}$ = ~($\overline{\text{DMAONS}}$ & OBJBUFINIT & CLK21' |
BEFLAG & DMA & P1H & $\overline{\text{DMAONS}}$ & CLK21' & (8H & ~HFLIP | ~8H & HFLIP) |
~UWR & ~DMAINIT)

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG & DMA
3. ~DMAINIT

Ulf Skutnabba, twitter: @skutis77

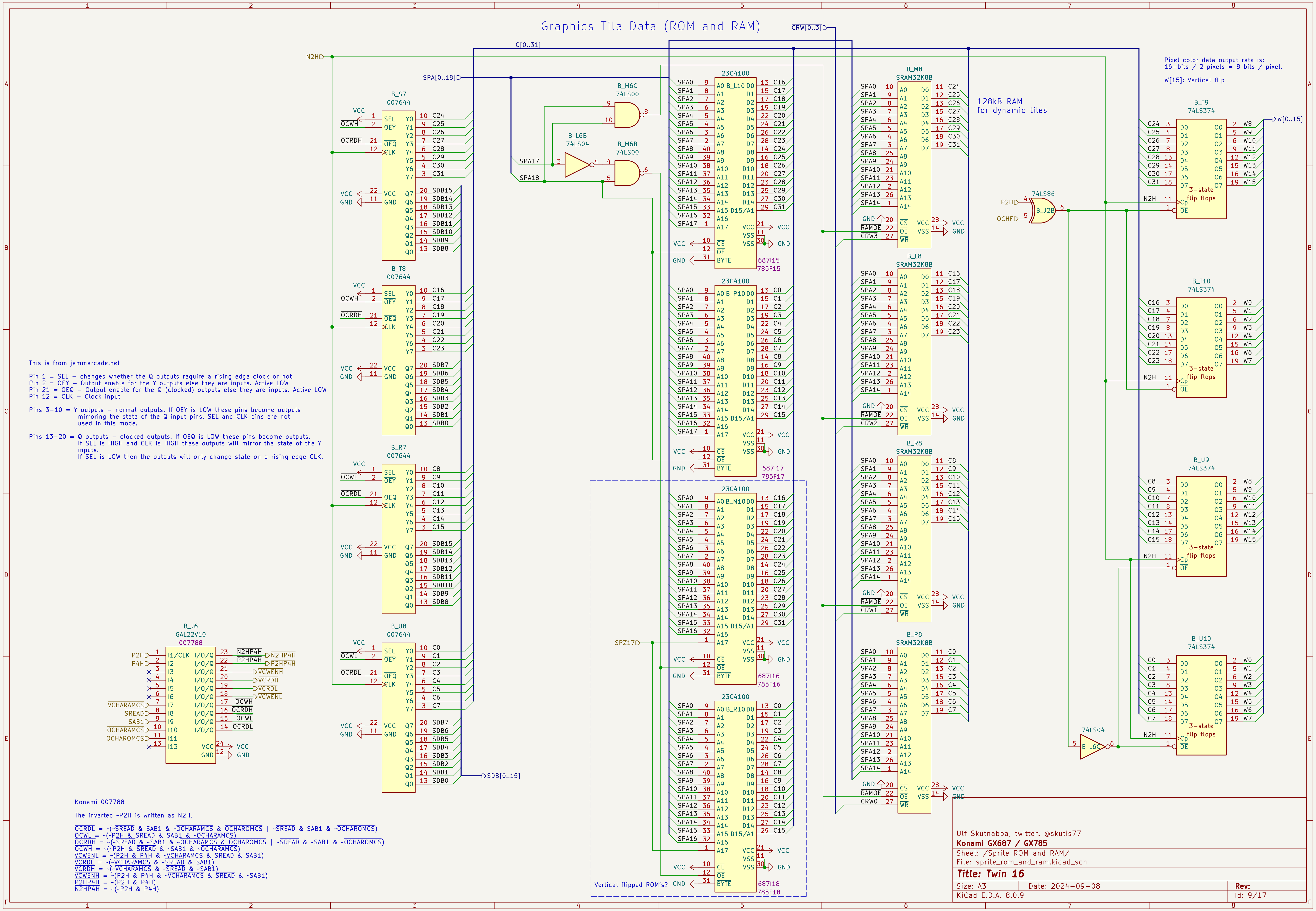
Konami GX687 / GX785

Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16

Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 8/17

Graphics Tile Data (ROM and RAM)



This is from jammarcade.net

Pin 1 = SEL - changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY - Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ - Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK - Clock input

Pins 3-10 = Y outputs - normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13-20 = Q outputs - clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.

Konami 007788

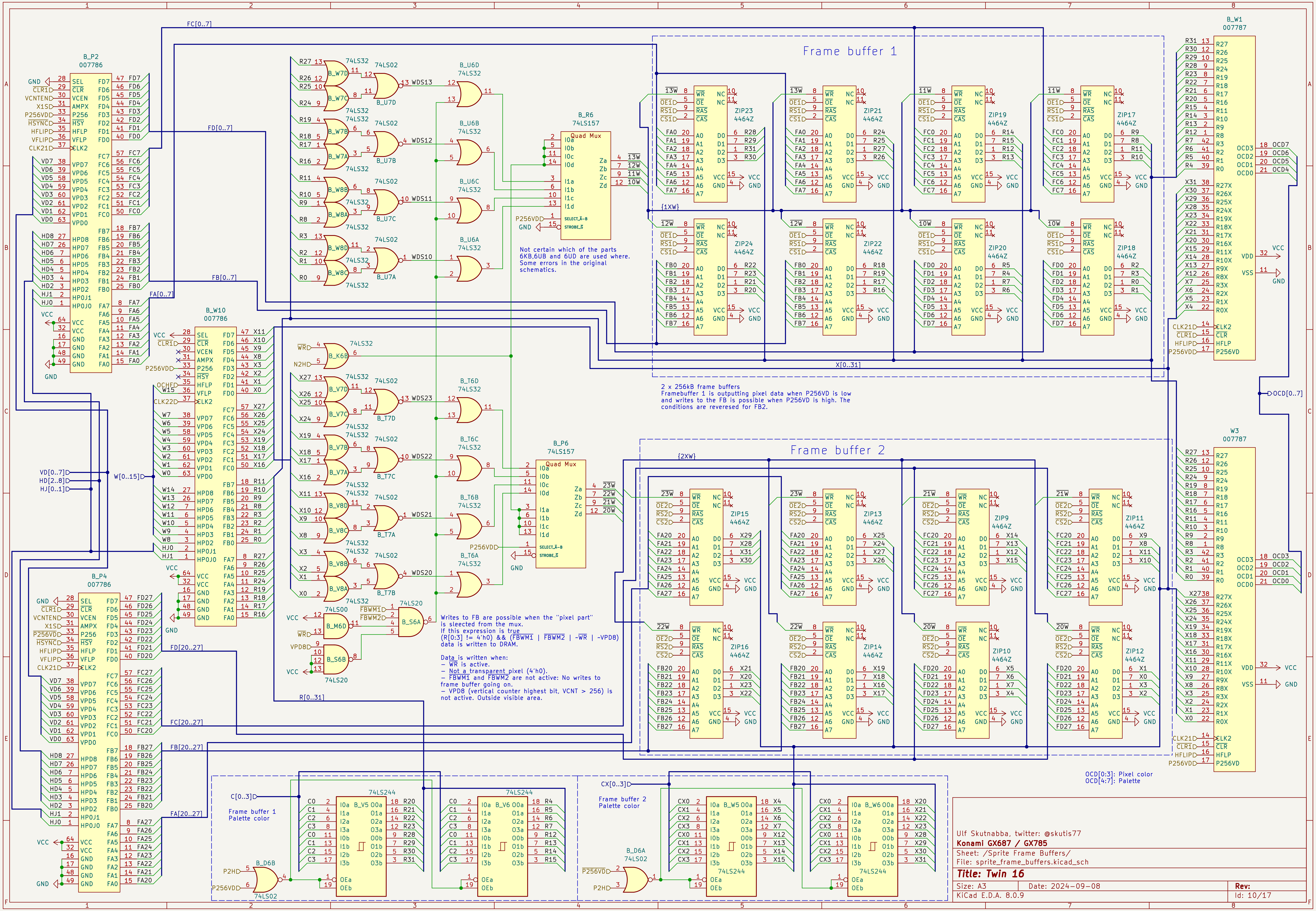
The inverted ~P2H is written as N2H.

OCRDL = ~(~SREAD & SAB1 & ~OCHARAMCS & OCHAROMCS | ~SREAD & SAB1 & ~OCHAROMCS)
OCWL = ~(~P2H & SREAD & SAB1 & ~OCHARAMCS)
OCRDH = ~(~SREAD & ~SAB1 & ~OCHARAMCS & OCHAROMCS | ~SREAD & ~SAB1 & ~OCHAROMCS)
OCWH = ~(~P2H & SREAD & ~SAB1 & ~OCHARAMCS)
VCWENL = ~(P2H & P4H & ~VCHARAMCS & SREAD & SAB1)
VCRDL = ~(~VCHARAMCS & ~SREAD & SAB1)
VCRDH = ~(~VCHARAMCS & ~SREAD & ~SAB1)
VCWENH = ~(P2H & P4H & ~VCHARAMCS & SREAD & ~SAB1)
P2HP4H = ~(P2H & P4H)
N2HP4H = ~(~P2H & P4H)

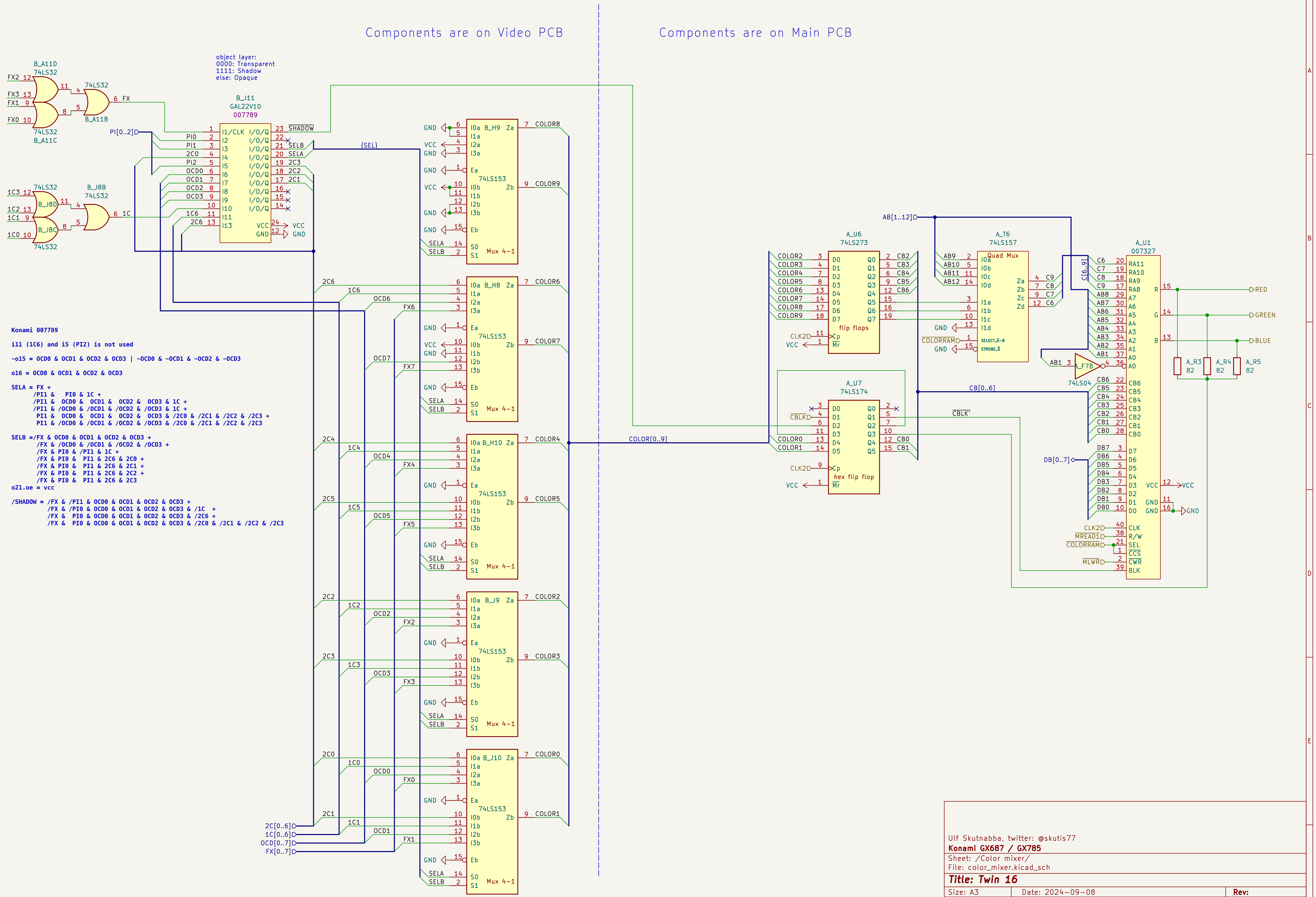
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Sprite ROM and RAM/
File: sprite_rom_and_ram.kicad_sch

Title: Twin 16

Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 9/17



Frame buffer 1
Frame buffer 2
Not certain which of the parts 6KB,6UB and 6UD are used where. Some errors in the original schematics.
2 x 256kB frame buffers
Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are revereseed for FB2.
Writes to FB are possible when the "pixel part" is slected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | -WR | -VPD8)
data is written to DRAM.
Data is written when:
- WR is active.
- Not a transparent pixel (4'h0).
- FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
- VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.
Frame buffer 1 Palette color
Frame buffer 2 Palette color
OCD[0:3]: Pixel color
OCD[4:7]: Palette
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Sprite Frame Buffers/
File: sprite_frame_buffers.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.9
Id: 10/17



Konami 007789

i11 (1C6) and i5 (PI2) is not used

```
~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = FX +
      /PI1 &   PI0 & 1C +
      /PI1 &  OCD0 &  OCD1 &  OCD2 &  OCD3 & 1C +
      /PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & 1C +
       PI1 &  OCD0 &  OCD1 &  OCD2 &  OCD3 & /2C0 & /2C1 & /2C2 & /2C3 +
       PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & /2C0 & /2C1 & /2C2 & /2C3

SELB =/FX & OCD0 & OCD1 & OCD2 & OCD3 +
       /FX & /OCD0 & /OCD1 & /OCD2 & /OCD3 +
       /FX & PI0 & /PI1 & 1C +
       /FX & PI0 &  PI1 & 2C6 & 2C0 +
       /FX & PI0 &  PI1 & 2C6 & 2C1 +
       /FX & PI0 &  PI1 & 2C6 & 2C2 +
       /FX & PI0 &  PI1 & 2C6 & 2C3
o21.oe = vcc

/SHADOW = /FX & /PI1 & OCD0 & OCD1 & OCD2 & OCD3 +
          /FX & /PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /1C  +
          /FX &  PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C6 +
          /FX &  PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C0 & /2C1 & /2C2 & /2C3
```

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

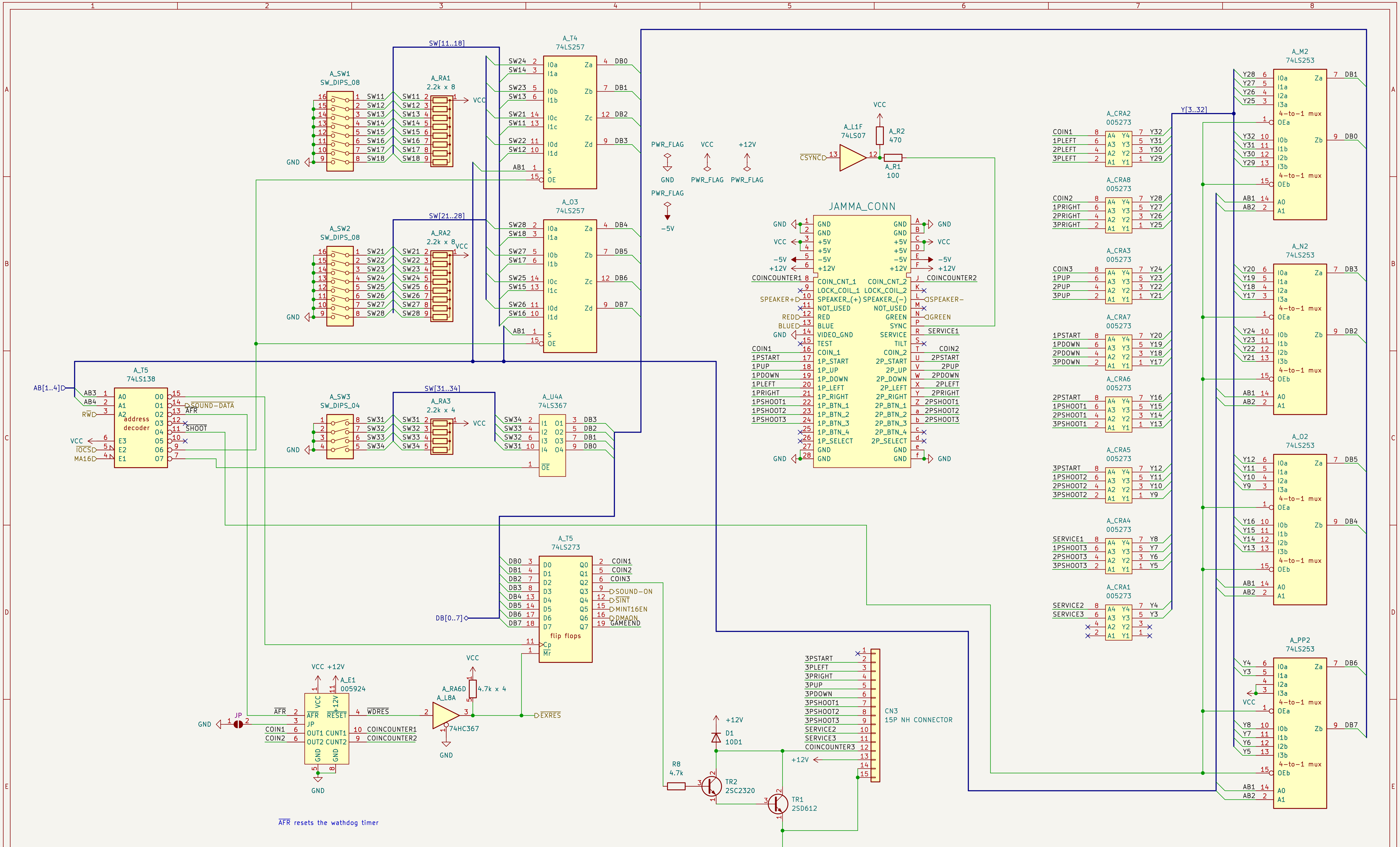
Sheet: /Color mixer/

File: color_mixer.kicad_sch

Title: Twin 16

Size: A3 | Date: 2024-09-08 | Rev:

KiCad E.D.A. 8.0.9 | Id: 11/17



Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

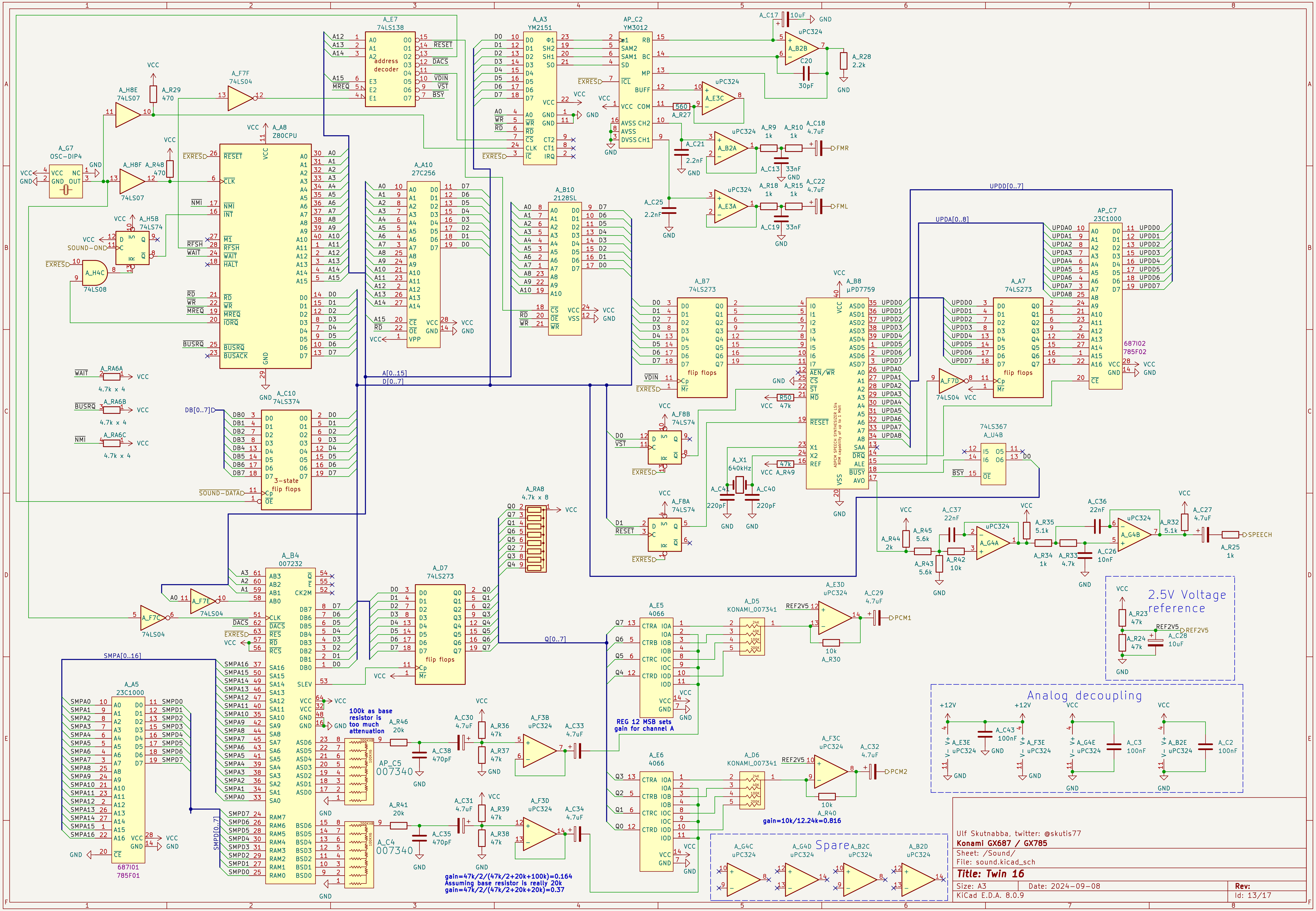
Sheet: /IO/

File: io.kicad_sch

Title: Twin 16

Size: A3 Date: 2024-09-08 Rev:

KiCad E.D.A. 8.0.9 Id: 12/17



Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

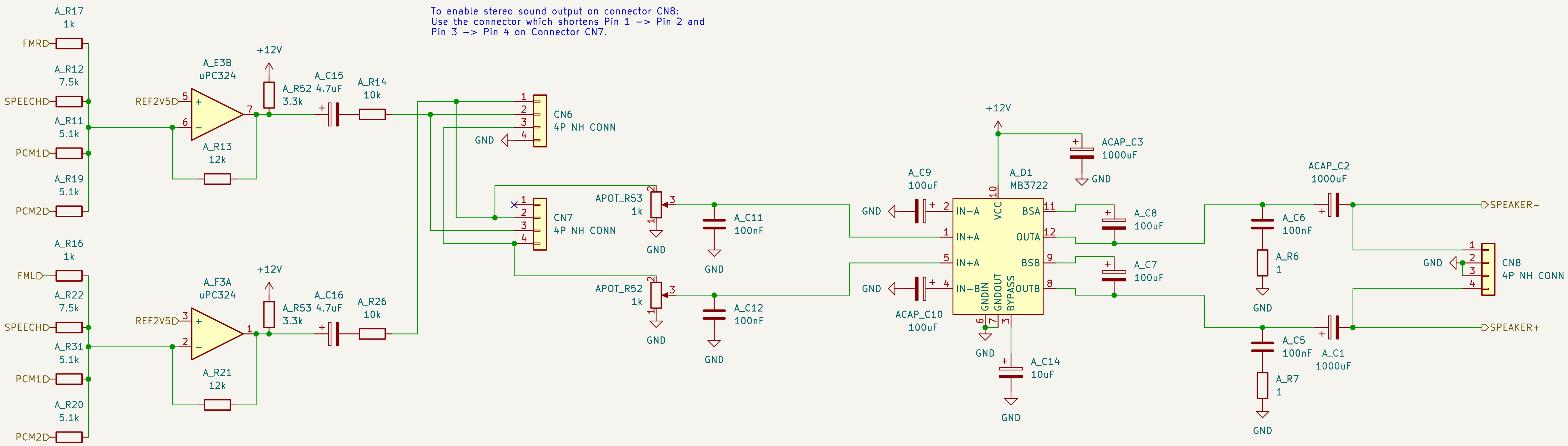
Sheet: /Sound/

File: sound.kicad_sch

Title: Twin 16

Size: A3 Date: 2024-09-08 Rev:

KiCad E.D.A. 8.0.9 Id: 13/17



Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Sound mixer/

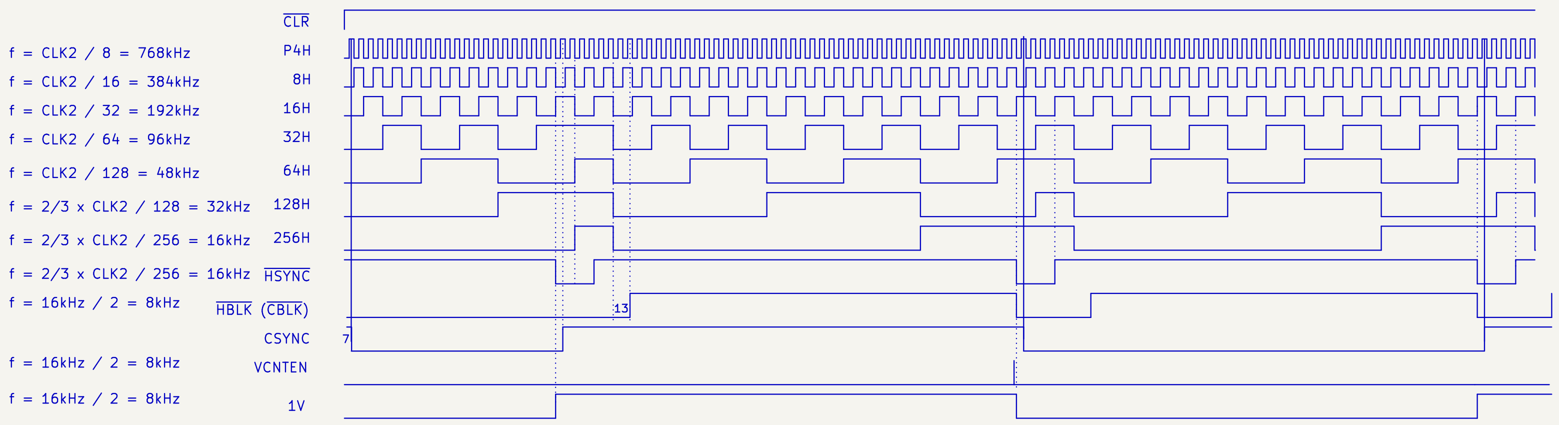
File: sound_mixer.kicad_sch

Title: Twin 16

Size: A3 | Date: 2024-09-08 | **Rev:**

KiCad E.D.A. 8.0.9 | Id: 14/17

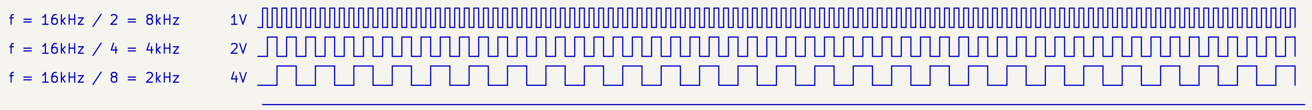
Horizontal signals

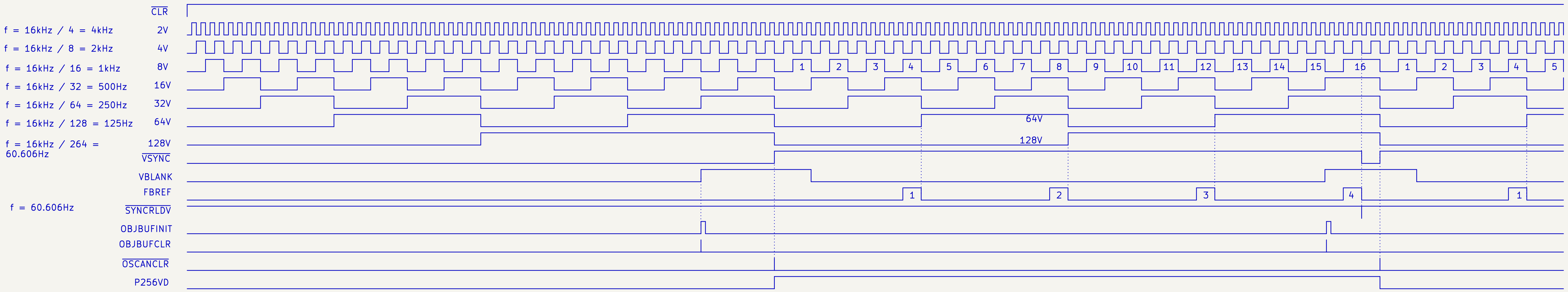


- The first VCNTEN is skipped after reset. It goes low 140ns before HSYNC goes low, and high again when HSYNC goes low. VCNTEN is active right before every second falling edge of HSYNC.
- CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals





Every 16th pulse is long.
Every 8th pulse is long.
Every 4th pulse is long.
Every 2nd pulse is long.
High period is longer

VBLANK High: 7 x 16 x 2 = 224, Low: 5 x 4 x 2 = 40

SYNCRLDV goes low 140ns before FBREF goes low.
SYNCRLDV goes high again when FBREF goes low.

OBJBUFCLR goes low one pixel clock before the rising edge of VBLANK and high again when VBLANK goes high.

60.606Hz, OSCANCLR goes low for 140ns - One pixel clock cycle.

P256VD goes high/low one P4H cycle (8 pixels) after VSYNC goes high.

FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

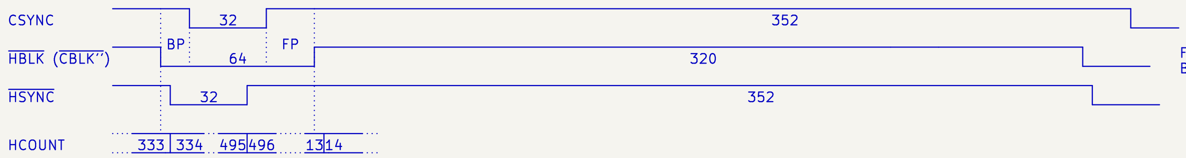
Horizontal and vertical synch timing diagrams

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

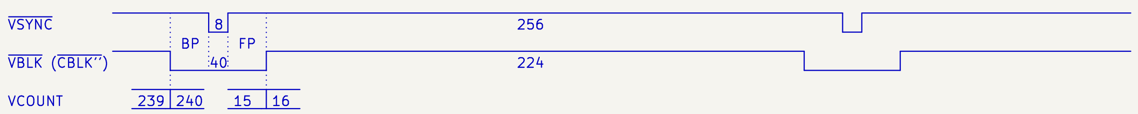
Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK" is at the output of color mixer 007327.
CBLK' is shifted inside the 007327 one pixel clock.

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

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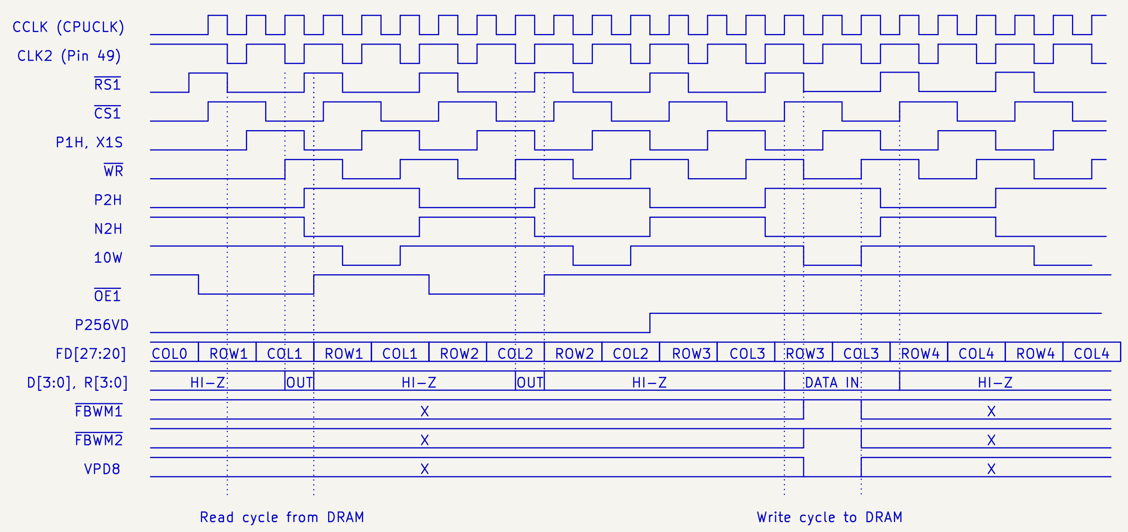

The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

Ulf Skutnabba, twitter: @skutis77

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Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch

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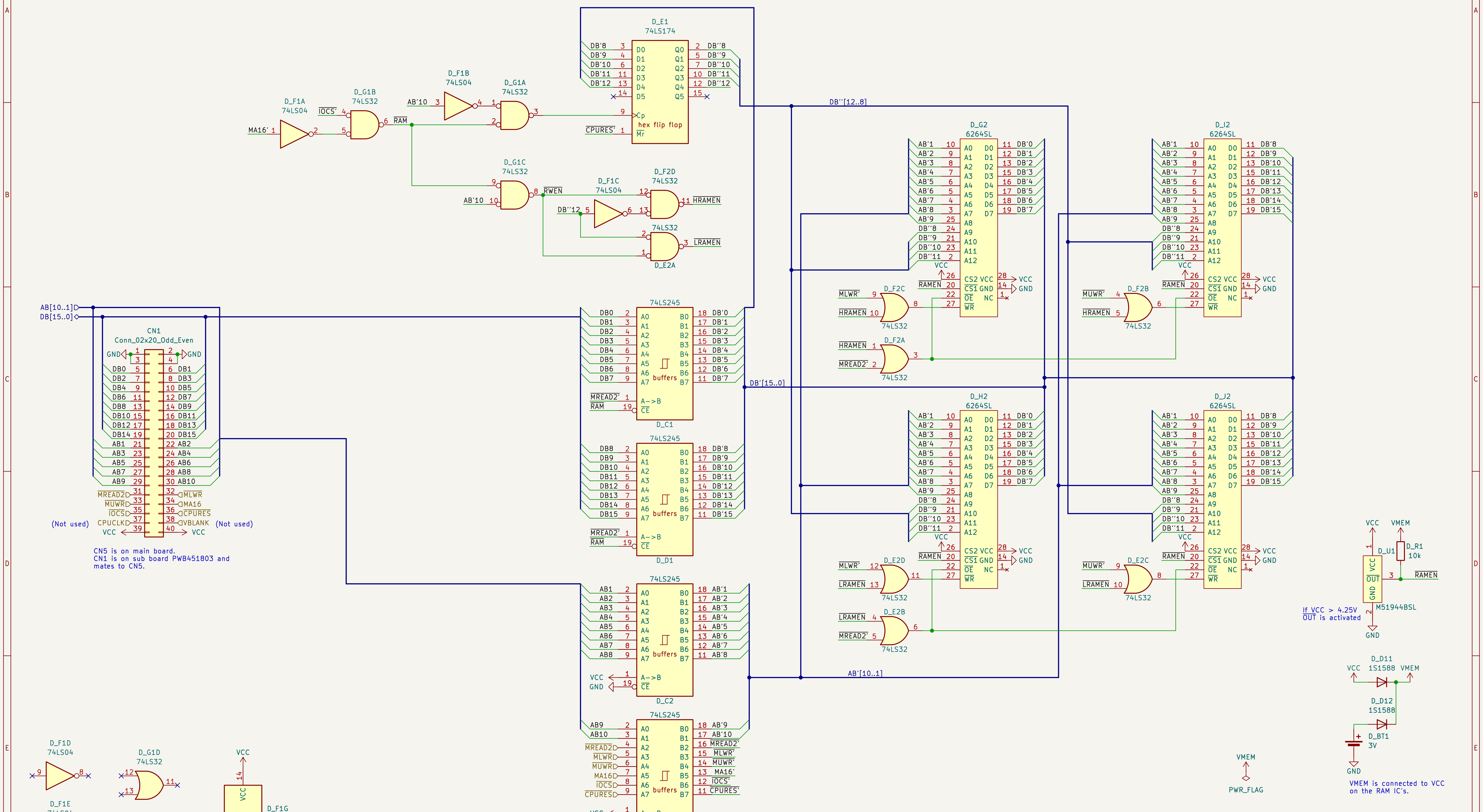

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